Phonetic Guided Speech Generation

Explicit Phoneme Alignment and Learned Acoustic Generation for Text-to-Speech

Course Project Report
Advanced Machine Learning & Deep Learning

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Code, results and audio samples: github.com/shiavm006/phonetic-guided-tts

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Abstract

Text-to-speech has two separable parts: deciding *when* each sound occurs (timing) and deciding *how* it sounds (timbre, coarticulation, prosody). We build a small, reproducible experiment on the single-speaker LJSpeech corpus that measures how far explicit phoneme alignment alone can go, and what a learned acoustic model adds. Using the Montreal Forced Aligner, we train two classical models, a LightGBM phoneme duration predictor (18.3 ms mean absolute error on speech sounds) and a LightGBM pause classifier (F1 64.2% versus 58.7% for a punctuation rule), and render their timing either with a duration-only baseline (one averaged mel frame per phoneme) or with a pretrained FastSpeech2 decoder conditioned on our durations. On 100 test sentences held out from every model, including the pretrained ones, the duration-only baseline reaches 36.6% word error rate (WER, Whisper), while the same text-only timing rendered by the neural decoder reaches 3.1%, close to real recordings (2.0%). Our explicit timing also beats FastSpeech2's own duration predictor (3.1% versus 5.3% WER) at statistically indistinguishable naturalness (UTMOS 3.70 versus 3.74). Fine-tuning the decoder on our alignments lowered its reconstruction loss by 18% but reduced naturalness by 0.43 UTMOS, a clear case of over-smoothing. A blind listening test (7 listeners, 70 judgements) preferred the neural system in 96% of trials. Diffusion decoding, an optional extension in the project brief, was not implemented; our results identify over-smoothing as the specific problem it would address.

1. Introduction

Classical statistical speech synthesis treated duration as an explicit, separately modelled quantity, while modern neural systems either learn alignment implicitly (Tacotron 2) or predict durations inside the network (FastSpeech 2). This raises a practical question for a hybrid design: if an aligner tells us precisely when every phoneme happens, how much of natural speech does that timing already give us, and which part genuinely requires a learned acoustic generator? The project brief frames this as a gap: *alignment alone specifies when a phoneme occurs but does not generate natural timbre, coarticulation or prosody.*

We answer the question by holding the timing fixed and swapping the sound generator, and vice versa. Our contributions are:

- An **explicit timing model** built from forced alignments: a duration predictor and a text-only pause classifier, both beating simple baselines.
- A **controlled comparison** in which the same timing plan is rendered by a duration-only baseline and by a neural decoder, isolating the contribution of learned acoustics (36.6% to 3.1% WER).
- Evidence that **explicit timing improves a neural TTS model**: replacing FastSpeech2's internal durations with ours lowers WER from 5.3% to 3.1% with no loss of naturalness.
- A **leak-free evaluation**: we reproduce the data split used to train the pretrained models and test only on sentences none of the models has trained on, with paired bootstrap confidence intervals throughout.
- A **fine-tuning study** that confirmed a predicted effect (handling raw alignment timing) but also exposed over-smoothing (lower loss, less natural speech).

2. Related Work

Forced alignment. The Montreal Forced Aligner (McAuliffe et al., 2017) trains speaker-adapted HMM-GMM acoustic models with Kaldi to place phoneme boundaries given a transcript; its pretrained English ARPAbet models are widely used to derive duration

targets for TTS. **Statistical parametric synthesis** (Zen, Tokuda & Black, 2009) modelled duration and acoustics with separate HMM-based components, which is the conceptual ancestor of our baseline.

Neural TTS. Tacotron 2 (Shen et al., 2018) learns alignment implicitly through attention, which can skip or repeat words. FastSpeech (Ren et al., 2019) and FastSpeech 2 (Ren et al., 2020) are non-autoregressive models with an explicit duration predictor trained on forced-alignment targets and a variance adaptor for pitch and energy; FastSpeech 2 is the closest relative of our hybrid. Glow-TTS (Kim et al., 2020) and Badlani et al. (2022) learn alignments without an external aligner. Grad-TTS (Popov et al., 2021) replaces the regression decoder with a diffusion model to reduce over-smoothing, the exact weakness we observe after fine-tuning. HiFi-GAN (Kong et al., 2020) converts mel spectrograms to waveforms and is shared by all our systems.

Evaluation. Intelligibility is commonly measured with the WER of a held-out speech recogniser; we use Whisper (Radford et al., 2023). For naturalness we use UTMOS (Saeki et al., 2022), a learned predictor of human mean opinion scores, alongside a small human preference study.

3. Data

Property	Value
Corpus	LJSpeech 1.1: one female speaker reading non-fiction, public domain
Size	13,100 clips, 23.92 hours, 22,050 Hz; clips 1.1–10.1 s (mean 6.6 s)
Transcripts	normalised column of metadata.csv (numbers and abbreviations written out)
Forced alignment	MFA, english_us_arpa acoustic model and dictionary; 13,100/13,100 aligned, 0 failures, 31.5 min
Features	80-band log-mel spectrograms, FFT 1024, hop 256 (11.6 ms per frame), 0–8 kHz, cached as float16
Our split	90/10 by utterance: 11,790 training and 1,310 test sentences
Leak-free evaluation set	100 sentences in our test split and in SpeechBrain's held-out set (see below)

Table 1. Dataset and preprocessing.

The mel frame counts derived from the alignments matched the cached spectrogram lengths exactly for every clip, so phoneme durations and acoustic frames line up without correction. Because LJSpeech has a single speaker, speaker-disjoint evaluation is impossible; all splits are utterance-disjoint.

Leak-free evaluation. The pretrained FastSpeech2 and HiFi-GAN checkpoints were trained by SpeechBrain on a 90/10 split of LJSpeech with seed 1234. Evaluating on sentences they had seen would flatter the neural systems, so we downloaded SpeechBrain's own preparation script, ran its split function with the same seed, and recovered their 1,327 held-out sentences. Our evaluation set is a random sample of 100 sentences from the intersection of their held-out set and our test split: no model, pretrained or ours, was trained on them.

4. Method

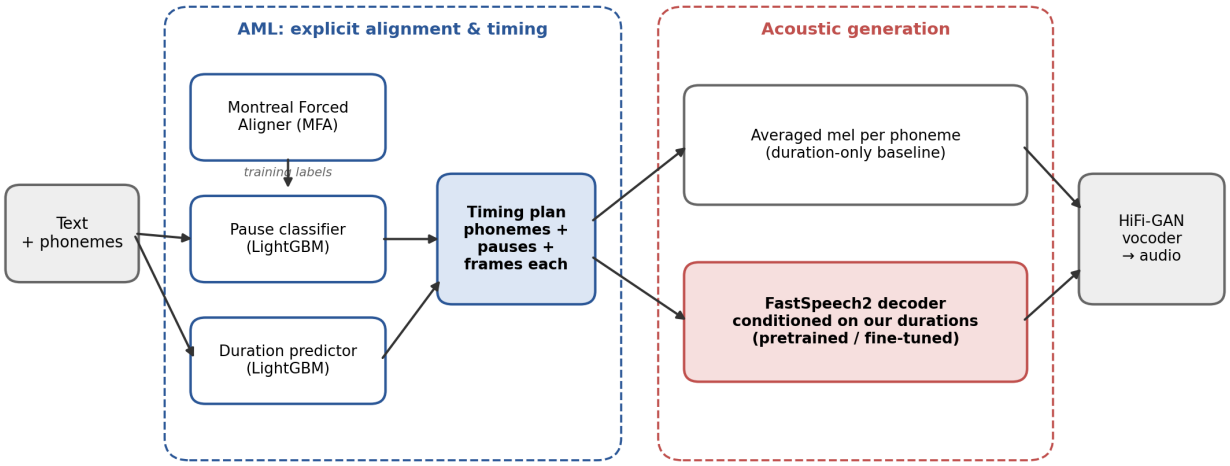


Figure 1. Pipeline. Forced alignments provide training labels; at synthesis time the timing plan is predicted from text and rendered either by averaged sounds or by a neural decoder; all systems share the same vocoder.

4.1 Duration predictor (AML)

Each phoneme is one training example whose target is its duration in mel frames. Features are the phoneme identity (stress removed), the previous and next phoneme, lexical stress, position within the word, word length, whether a pause follows, relative position in the sentence and sentence length. We train a LightGBM regressor (600 trees, 63 leaves, learning rate 0.05) and compare it with the simplest baseline, the mean training duration of each phoneme. Leading and trailing silences are excluded because their length is arbitrary.

4.2 Pause classifier (AML)

A text-only system must also decide *where* the speaker pauses. For every gap between two words we label whether MFA found a silence of at least 100 ms; shorter silences are mostly stop closures rather than phrase breaks. Features are the punctuation after the word (aligned from the transcript with sequence matching, 99.4% word match rate), the current and next word when they are among the 80 most frequent words, word lengths, position, words remaining, and words and characters since the last punctuation mark. A LightGBM classifier is trained on the training split, with its decision threshold (0.35) tuned for F1 on a held-out 10% of the training sentences, and compared with the rule *pause if and only if there is punctuation*.

4.3 Duration-only baseline

For each phoneme we average every mel frame labelled with it across the training set (41 units: 39 ARPAbet phonemes, silence and unknown-word). A sentence is synthesised by repeating each phoneme's average frame for its duration and passing the result to the vocoder. This system has perfect access to timing but no model of how sounds change over time.

4.4 Hybrid: neural decoder conditioned on our timing (DL)

We use the SpeechBrain FastSpeech2 checkpoint for LJSpeech but bypass its duration predictor: the encoded phonemes are expanded according to *our* durations and decoded into a mel spectrogram, so the output length equals the sum of our durations exactly. Pauses are passed as the model's `spn` token. Its vocabulary contains no word-separator token; our first experiments inserted spaces between words as in the model's documentation, which were silently mapped to `<unk>` and inflated its WER from 6.9% to 20.8% on our initial test set. All

reported results use the corrected format, and our code now rejects unknown tokens.

4.5 Decoder fine-tuning (DL)

To adapt the decoder to our alignments, we fine-tuned its decoder, output projection and postnet (22.5M of 43.3M parameters) on our training split with our MFA durations and cached mels as targets, keeping the encoder and variance predictors frozen. Training used an L1 loss on both decoder and postnet outputs, AdamW (learning rate 5e-5), batch size 16, mixed precision, and six epochs (about 2.3 minutes each on a T4 GPU), keeping the epoch with the lowest L1 on 200 held-out training sentences.

4.6 Systems compared

System	Pause placement	Durations	Sound generator
original	real recording	real	real recording
ceiling	real	real	real mel → HiFi-GAN
baseline_true / _pred / _text	MFA / MFA / predicted	MFA / LightGBM / LightGBM	averaged mels
hybrid_true / _pred / _text	MFA / MFA / predicted	MFA / LightGBM / LightGBM	FastSpeech2 decoder
fastspeech2	predicted	FastSpeech2's own	FastSpeech2 decoder
*_ft	as above	as above	fine-tuned decoder

Table 2. Systems. Suffixes _true and _pred use pause locations from the reference alignment; _text systems use only the text and are the deployable configuration.

5. Evaluation Protocol

- **Intelligibility:** WER of Whisper small.en on each system's audio against the reference transcript, both normalised with Whisper's English text normaliser.
- **Uncertainty:** 95% bootstrap confidence intervals from 5,000 resamples of the 100 sentences. Comparisons are *paired*: every system is resampled on the same sentences, so sentence difficulty cancels out. A difference is called significant when its interval excludes zero.
- **Duration accuracy:** mean absolute error against MFA durations; for the comparison with FastSpeech2 both predictors receive identical phoneme sequences.
- **Naturalness:** UTMOS (utmos22_strong), an automatic estimate of human mean opinion score on a 1–5 scale.
- **Human preference:** a blind A/B test with randomised order (Section 6.5).

6. Results

6.1 Explicit timing models

Duration model	MAE, all	MAE, speech sounds	MAE, pauses
Per-phoneme mean	33.0 ms	28.8 ms	145.6 ms
LightGBM (ours)	**21.6 ms**	**18.3 ms**	**112.9 ms**

Table 3. Duration prediction on our test split (1,310 sentences).

Context reduces the error by about a third, to under two mel frames on speech sounds. By gain importance, phoneme identity dominates, followed by the next and previous phoneme, sentence position and the presence of a following pause, which captures phrase-final lengthening. Pauses remain the hardest case: their length depends on meaning and phrasing

that these features cannot see. On the 100 evaluation sentences, with identical inputs, LightGBM is also more accurate than FastSpeech2's internal predictor (17.8 ms versus 23.2 ms, Figure 2). Part of this advantage is that LightGBM is trained on the same alignments it is evaluated against.

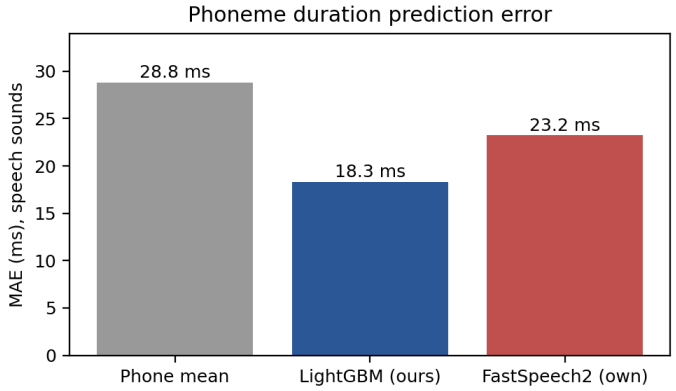


Figure 2. Duration error on speech sounds for the per-phoneme mean, our LightGBM model and FastSpeech2's own predictor.

Pause model	Accuracy	Precision	Recall	F1
Rule: pause iff punctuation	93.5%	68.2%	51.5%	58.7%
LightGBM classifier (ours)	**93.9%**	**68.2%**	**60.6%**	**64.2%**

Table 4. Pause placement on our test split; 9.2% of word gaps contain a pause of at least 100 ms.

The classifier finds nine percentage points more real pauses at the same precision. Pause placement is genuinely hard to predict from text: only about 56% of long pauses (over 200 ms) follow punctuation, because a reader also breathes inside long unpunctuated phrases, often before words such as *and*, *which* or *of*.

Model comparison, tuning and ablation

To justify the choice of LightGBM we trained four alternative model families on the same features and split. Random forest and the MLP were trained on a 200,000-row subsample to keep CPU time manageable, which may understate them slightly.

Duration model	MAE, all	MAE, speech	MAE, pauses	Gap to LightGBM (95% CI)
Per-phoneme mean	33.0 ms	28.8 ms	145.6 ms	+11.2 to +11.5 ms
Ridge regression (linear)	29.3 ms	25.1 ms	144.3 ms	+7.5 to +7.8 ms
Random forest	22.3 ms	18.8 ms	118.1 ms	+0.6 to +0.8 ms
MLP (128-64)	22.7 ms	19.3 ms	115.2 ms	+1.0 to +1.2 ms
LightGBM	**21.6 ms**	**18.3 ms**	**112.9 ms**	**best**

Table 4a. Duration models on our test split; intervals are paired bootstraps over test sentences.

LightGBM is significantly better than every alternative. The linear model is 7.7 ms worse, which shows that duration depends on interactions between features, for example a vowel lengthening more when it is both stressed and phrase-final. Three-fold cross-validation by sentence over tree size and learning rate found 127 leaves slightly better than our 63 (22.3 versus 22.6 ms), a 0.3 ms gain that is small next to the 7.7 ms gained by choosing a non-linear model.

Feature group removed	MAE, all	Increase
None (full model)	21.6 ms	–
Neighbouring phonemes	27.3 ms	+5.7 ms
Phoneme identity	26.6 ms	+5.0 ms
Position in word and sentence	22.8 ms	+1.2 ms
Stress	21.9 ms	+0.3 ms
Following pause	21.7 ms	+0.0 ms

Table 4b. Feature ablation: LightGBM retrained without each feature group.

The surrounding sounds matter slightly more than the sound itself: timing reflects coarticulation. The following-pause flag adds nothing on its own because the next-phoneme feature already encodes a following silence; the ablation measures unique contribution, which gain importance cannot show. By sound class, nasals are predicted best (16.0 ms) and vowels worst among speech sounds (19.9 ms), while pauses improve least over the per-phoneme mean (22% versus 35–43%).

Pause model	Precision	Recall	F1
Rule: pause iff punctuation	68.2%	51.5%	58.7%
Logistic regression	61.0%	61.2%	61.1%
Random forest	68.4%	58.4%	63.0%
MLP (64-32)	66.4%	60.7%	63.5%
LightGBM	**68.2%**	**60.6%**	**64.2%**

Table 4c. Pause models; each threshold tuned for F1 on held-out training sentences.

Every learned model beats the punctuation rule, and the three non-linear models lie within about one F1 point of each other, with LightGBM highest. We did not compute confidence intervals for this task, so we treat the non-linear models as comparable.

6.2 Intelligibility

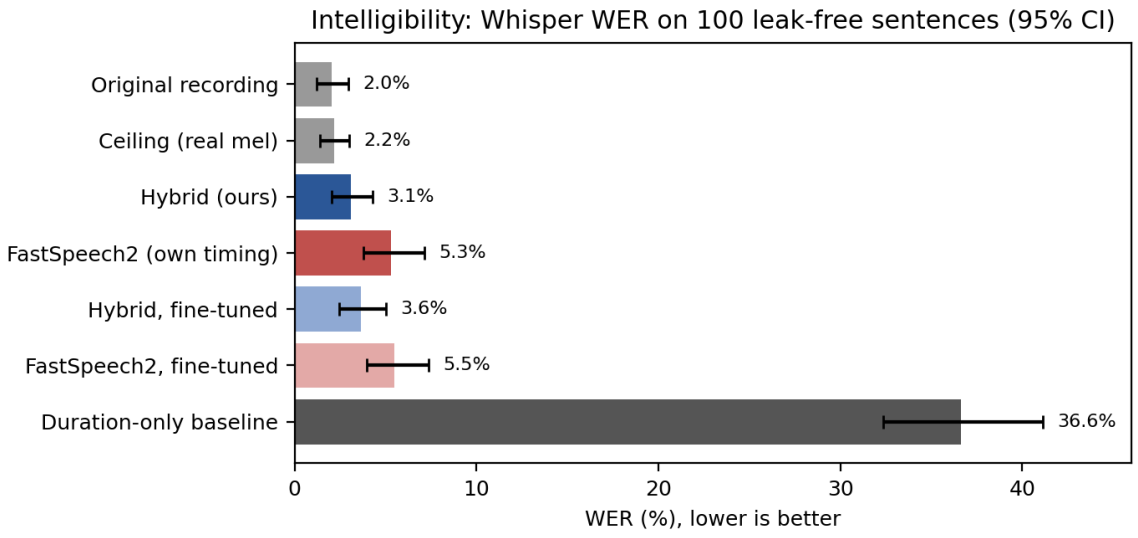


Figure 3. Whisper WER on the 100 leak-free sentences with 95% bootstrap confidence intervals.

System	Decoder	WER (95% CI)	UTMOS
original recording		2.0% [1.2, 3.0]	4.37
ceiling (real mel)		2.2% [1.4, 3.0]	4.08
hybrid_text (ours, text-only)	**pretrained**	**3.1% [2.0, 4.3]**	**3.70**
hybrid_pred	pretrained	2.9% [1.9, 4.1]	–
hybrid_true	pretrained	4.2% [3.0, 5.7]	–
fastspeech2 (own durations)	pretrained	5.3% [3.8, 7.1]	3.74
hybrid_text_ft	fine-tuned	3.6% [2.5, 5.0]	3.27
hybrid_true_ft	fine-tuned	2.8% [1.9, 3.8]	3.17
fastspeech2_ft	fine-tuned	5.5% [4.0, 7.4]	3.28
baseline_text	averaged mels	36.6% [32.3, 41.1]	1.24
baseline_true / baseline_pred	averaged mels	37.1% / 36.1%	–

Table 5. Intelligibility and naturalness of all systems on the leak-free evaluation set.

Comparison (same sentences)	Difference, 95% CI	Interpretation
hybrid_text − baseline_text (WER)	−37.9 to −29.4 points	learned acoustics are essential
hybrid_text − fastspeech2 (WER)	−3.6 to −0.9 points	our timing beats the model's own
hybrid_text − fastspeech2 (UTMOS)	−0.09 to +0.03	no difference in naturalness
ceiling − original (WER)	−0.3 to +0.5 points	the vocoder costs no intelligibility

Table 6. Key paired comparisons.

Three findings follow. **First, learned acoustics are the dominant factor.** With identical text-only timing, averaged sounds give 36.6% WER and the neural decoder 3.1%, roughly twelve times fewer errors. The baseline stays near 37% whichever timing it receives (true, predicted or text-only), so timing is not its bottleneck. **Second, explicit timing improves the neural model.** Given the same phonemes and pauses, using our durations instead of FastSpeech2's lowers WER from 5.3% to 3.1%, and the improvement is significant. **Third, the text-only hybrid is close to real speech:** 30 of the 100 sentences contain any error at all, against 43 for FastSpeech2 and 95 for the baseline.

6.3 Naturalness

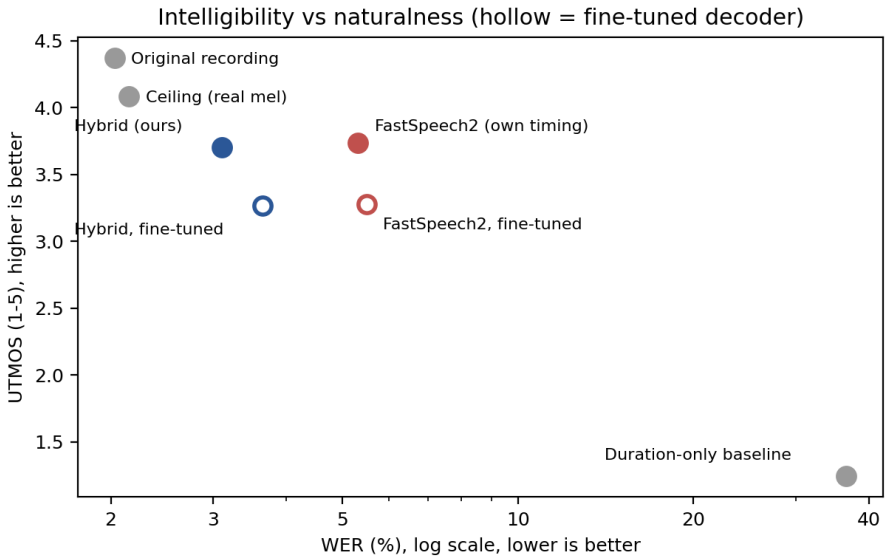


Figure 4. Intelligibility against naturalness. Filled markers use the pretrained decoder, hollow markers the fine-tuned one.

Naturalness and intelligibility do not move together (Figure 4). The hybrid and FastSpeech2 are equally natural (3.70 versus 3.74, difference not significant), so the hybrid's intelligibility gain comes at no cost. The remaining gap to real speech is in naturalness rather than intelligibility: the hybrid is about one WER point from the recordings but 0.38 UTMOS below the vocoder ceiling. The vocoder itself costs naturalness (4.37 to 4.08) but not intelligibility.

6.4 Decoder fine-tuning: lower loss, worse sound

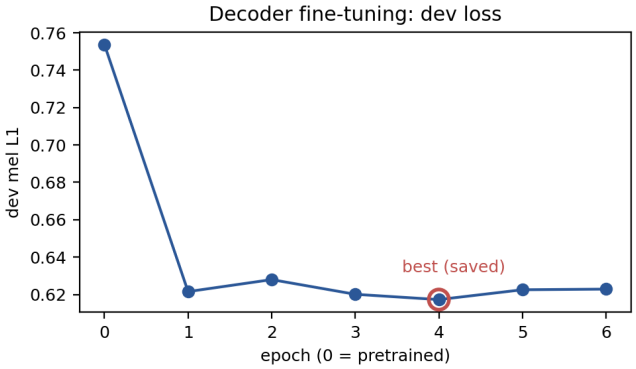


Figure 5. Development L1 during fine-tuning (epoch 0 is the pretrained decoder).

The pretrained decoder rendered raw MFA timing worse than predicted timing (hybrid_true 4.2% versus hybrid_pred 2.9% WER). We traced this to ultra-short phonemes: 3.0% of MFA durations are two frames or fewer, against 0.2% of LightGBM's. Our hypothesis was that fine-tuning on our alignments would teach the decoder to render such timing. It did: hybrid_true improved from 4.2% to 2.8% WER (paired CI −2.5 to −0.6 points), and the development L1 fell from 0.754 to 0.617, with most of the gain in the first epoch and mild overfitting after the fourth.

However, fine-tuning **reduced naturalness for every system by about 0.43 UTMOS** (hybrid_text 3.70 to 3.27, CI −0.48 to −0.38) and slightly increased the text-only WER (3.1% to 3.6%). Minimising an average per-frame error pulls the decoder towards an average spectrogram, which blurs fine spectral detail: lower loss, flatter speech. This is the over-smoothing problem that motivates diffusion decoders such as Grad-TTS. Our recommended system is therefore the **pretrained decoder driven by our predicted timing** (hybrid_text).

6.5 Human listening test

Seven listeners (all on headphones; five native English speakers) judged ten sentences of varying length in a blind A/B test with randomised order, choosing the easier voice and rating each voice from 1 (no words understood) to 5 (everything understood). The neural system was preferred in 67 of 70 judgements, never lost (3 ties), and received a mean clarity of 4.11 against 1.69 for the baseline; the difference of 2.43 points has a 95% confidence interval of 1.56 to 3.06 when resampling listeners. The test was run before the input-format correction (Section 4.4), so it compares the baseline with a weaker neural system and is a conservative estimate. No listeners were available for a second test of the final systems; UTMOS serves as the naturalness measure for those.

6.6 What the spectrograms show

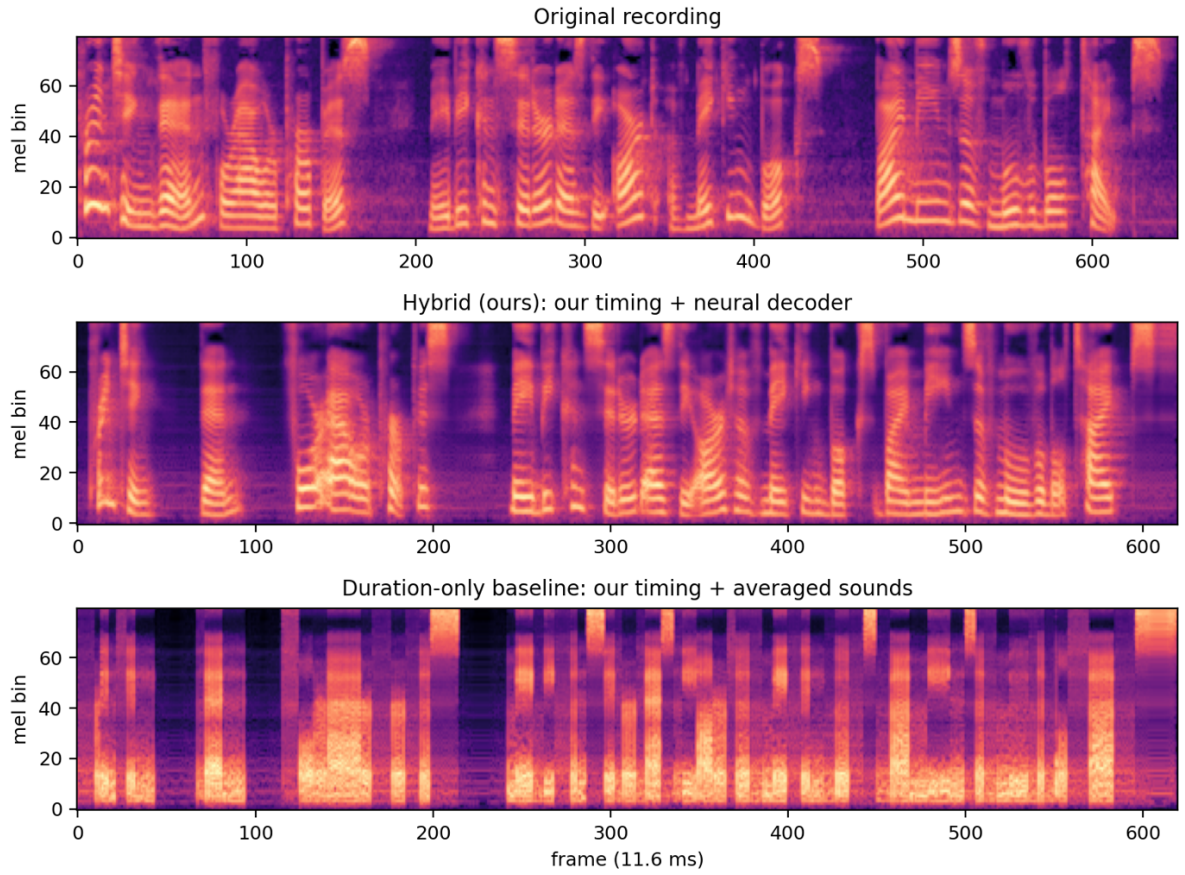


Figure 6. The same test sentence: real recording, our hybrid, and the duration-only baseline. Both synthetic versions use the same text-only timing.

The real recording and the hybrid both show curved horizontal stripes in voiced regions: harmonics of a moving pitch, which carry intonation, plus smooth transitions between sounds. The baseline has the same rhythm but consists of flat rectangular blocks with no harmonic structure. Averaging a vowel over many utterances spoken at different pitches cancels the harmonics into a smooth envelope, so the voice loses pitch entirely, and every phoneme boundary is an abrupt jump. The figure shows the brief's gap directly: correct timing, missing timbre, coarticulation and prosody.

7. Failure Analysis

System	Reference	Whisper transcript
hybrid_text	at walsall in staffordshire	at walsolinst afordshire
fastspeech2	at walsall in staffordshire	at wall salenced affords sure
hybrid_text	leon f czolgosz	leon f cholgo sh
hybrid_text	…a hogshead per week	…a hog is head per week
fastspeech2	…to carry gun lieutenant j c day end quote	…to carry the jc day
baseline_text	at walsall in staffordshire	that will still inspire for sure
baseline_text	such ideas of grandeur were apparently accompanied by notions of oppression	the shois as a friend we are within perfectly a company by motions of oppression

Table 7. Typical errors on the evaluation set.

- **Rare words and proper nouns** (Walsall, Staffordshire, Czolgosz) cause most neural errors. The pronunciation is usually close, but word boundaries blur and the recogniser

has no prior for the word.

- **Word boundaries inside compounds** (hogshead heard as *hog is head*) reflect weak coarticulation cues at joins.
- **Scoring artefacts:** some errors are spelling variants (*any one* versus *anyone*), so very low WERs overstate real differences slightly.
- **Baseline:** only rhythm and some stressed syllables survive (*oppression* is recovered). When the audio is mostly unintelligible, Whisper does not abstain but invents fluent unrelated text, producing WERs above 100% on short sentences. WER is therefore reliable for comparing good systems but overstates the randomness of a poor one, which is why the human test matters.
- **Pauses without punctuation** remain the main timing error (113 ms MAE on pause durations, 61% recall on pause placement).
- **Implementation pitfalls:** an input token that is valid in the documentation but absent from the model's vocabulary silently tripled the error rate; verifying inputs against the model vocabulary changed the conclusions.

8. Limitations

- Single speaker: speaker-disjoint evaluation is impossible; results may not transfer to multi-speaker or expressive speech.
- Text-only systems still use MFA's pronunciations, which come from a dictionary lookup that can be informed by the audio for words with several variants; a grapheme-to-phoneme model would make them fully text-only.
- The pretrained models selected their checkpoints on SpeechBrain's validation set, which contains our evaluation sentences (held out from training, not from model selection).
- The human test covered the earlier neural system; the final systems were compared by WER and UTMOS, which is an automatic estimate rather than a substitute for listeners.
- With 100 evaluation sentences, differences of about one WER point are near the resolution of the test.

9. Conclusion and Future Work

Explicit phoneme alignment decides *when* sounds occur; a learned acoustic model decides *how* they sound. With timing held fixed, replacing averaged sounds by a neural decoder cuts WER from 36.6% to 3.1%, so learned acoustics are indispensable. With the decoder held fixed, replacing its internal durations by our explicit duration and pause models lowers WER from 5.3% to 3.1% at equal naturalness, so explicit phonetic guidance still adds value to a neural system. The remaining gap to real speech is naturalness, and fine-tuning with a regression loss widened it. Natural next steps are a diffusion decoder (Grad-TTS) to counter over-smoothing, a grapheme-to-phoneme front end for unseen words, a human mean opinion score study on the final systems, and a multi-speaker extension on VCTK.

Scope: why no diffusion model

The brief lists a diffusion decoder as an optional advanced extension, to be attempted after a working baseline. We prioritised what the comparison depends on: a strong explicit-timing baseline, a neural decoder driven by that timing, a leak-free test set and paired statistics. The results now make the case for diffusion concrete rather than speculative. The text-only hybrid is within about one WER point of real speech, so intelligibility is largely solved, while its naturalness remains 0.38 UTMOS below the vocoder ceiling. Fine-tuning the regression decoder lowered its loss by 18% yet reduced naturalness by 0.43, the over-smoothing

behaviour that diffusion decoders such as Grad-TTS were designed to avoid. Because Grad-TTS also expands its encoder output by phoneme durations, our duration and pause models could drive it unchanged, which makes it the natural next experiment.

Reproducibility

All code is in one Colab notebook in the repository, which downloads LJSpeech and the pretrained models from official sources and the forced alignments and fine-tuned decoder from the repository's release. Every split and sampling step is seeded, and a complete run takes about 1.5 hours on a free Colab T4 GPU. Forced alignment requires a separate runtime because the conda installation it needs conflicts with PyTorch in the same Colab session. The repository also contains all result tables, figures, transcripts, the anonymised listening-test data and audio samples of every system.

Team Contributions

Member	Contribution
Shivam Mittal	Project lead: research questions and hypotheses, system design, hybrid system (FastSpeech2 decoder driven by our durations), leak-free evaluation design, GitHub repository and notebook
Divya Singh	Data and AML models: LJSpeech preparation, forced alignment, duration predictor, pause classifier, feature engineering and baselines
Sanath Waraikar	DL and synthesis: averaged-mel baseline, FastSpeech2 and HiFi-GAN integration, decoder fine-tuning, spectrogram analysis
Aryan	Evaluation: Whisper WER, UTMOS, bootstrap confidence intervals, listening test design and analysis, failure analysis, figures
Khushi Batra	Literature review (related work), proofreading of the report and presentation

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